MCS-103	ADVANCE COMPUTER NETWORKS	L T P 3 1 0	4 Credits

Course Objectives:

CO1	to study the problematic of service integration in TCP/IP networks focusing on protocol design, implementation and performance issues
CO2	to debate the current trends and leading research in the computer networking area.
CO3	to understand theoretical and practical concepts behind the design of multiconstained applications and services
CO4	to recognize the need for service integration and discuss how it can be accomplished
CO5	Design and implement a network protocol

MODULE I

IP addressing, subnetting, supernetting, variable length subnet masking (CIDR notation), ARP, RARP, ICMP,IGMP

MODULE II

IPv6, Next Generation IP protocol, Wireless Networks, Mobility in networks, Mobile IP, Mobile TCP, TCP extensions for high speed network, SCTP,

MODULE III

IP multicasting, Multicast routing TCP/IP programming. P2P file sharing, structure overlay network, Virtual Private N/W, Configuration of VLAN

References

1. *Larry Peterson and Bruce Davie, "*Computer Networks: A Systems Approach*", Third Edition, Morgan Kauffmann, 2003, ISBN: 1-55860-832-X.*
2. *W. Richard Stevens, Bill Fenner and Andrew Rudoff, "*UNIX Network Programming, Volume 1: Networking APIs - Sockets and XTI*", Third Eddition, Prentice Hall, 2004, ISBN: 0-13-141155-1.*
3. *Behrouz A. Forouzan "Data Communications and Networking", McGraw-Hill.*
4. *Behrouz A. Forouzan "TCP/IP" McGraw-Hill.*

Course Objectives:

CO1	to identify and discuss the concepts underlying IPv6 protocol, and their main characteristics and functionality
CO2	to understand the principles and functionality of mobile IP, explaining its concretization in IPv6; to understand the needs of optimization of the mobility mechanisms and description of some extensions that aim to reduce handover latency and requirements from terminals
CO3	to understand and explain the design issues in transport services in face of applications and services requirements
CO4	to discuss relevant management issues and devise adequate network management solutions
CO5	to identify and assess possible research opportunities and difficulties within the course scope.